Remark 3.15. By [19, Remark 2.28], if a definable function between two sets in Γ is a vrv-contraction then it is piecewise $\mathbb{Z}$-affine (with constant terms of the form $\mathrm{vrv}(t)$, where $t \in \mathrm{RV}$ is definable, that is, $t \in \mathrm{RV}(\mathbb{S})$). Moreover, a definable bijection between two objects of $\Gamma[k]$ is a $\Gamma[k]$-morphism if and only if it is definably a piecewise $\mathrm{GL}_k(\mathbb{Z})$-transformation. The "if" direction is clear. For the "only if" direction, see [8, Lemma 10.1] or [19, Lemma 2.29].

Resolved referee R050 [Clarification] – Piecewise integral-affine behavior.
Original source line 1289. The first sentence needs "piecewise $\mathbb{Z}$-affine," not globally "$\mathbb{Z}$-linear." For example, $\gamma \mapsto |\gamma|$ is the vrv-contraction of the definable map $t \mapsto t$ for $\mathrm{vrv}(t) \geq 0$, $t \mapsto t^{-1}$ otherwise. Its two integral-linear formulas differ. Retain the specified restriction on the translation constants.

Author resolution. The author approves the piecewise $\mathbb{Z}$-affine formulation. The restriction on translation constants and the following piecewise $\mathrm{GL}_k(\mathbb{Z})$ characterization are retained.

Definition 3.16 (Γ-categories with volume forms)**.** An object of the category $\mu\Gamma[k]$ is a definable pair (I, ω), where $I \in \Gamma[k]$ and $\omega : I \longrightarrow \Gamma$ is a function. A $\mu\Gamma[k]$-morphism between two objects (I, ω), (J, σ) is a $\Gamma[k]$-morphism $F : I \longrightarrow J$ such that for all $\alpha \in I$,

$$\omega(\alpha) = \sigma(F(\alpha)) + \mathrm{Jcb}_\Gamma F(\alpha).$$

The category $\mu\Gamma^{\mathrm{fin}}[k]$ is the obvious full subcategory of $\mu\Gamma[k]$.

Notation 3.17. For each $(U, f, \omega) \in \mu\mathrm{RV}[k]$, write $\omega_f : U \longrightarrow \Gamma$ for the function given by $u \longmapsto \Sigma(\mathrm{vrv} \circ f)(u) + \omega(u)$. Similarly, for $(I, \sigma) \in \mu\Gamma[k]$, write $\sigma_I : I \longrightarrow \Gamma$ for the function given by $\gamma \longmapsto \Sigma\gamma + \sigma(\gamma)$.

There is a natural map $\Gamma[*] \longrightarrow \mathrm{RV}[*]$ given by $I \longmapsto (I^\sharp, \mathrm{id})$ (see Notation 2.7). This map induces a commutative diagram in the category of graded semirings:

$$\begin{array}{ccc} \mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RES}[*] \\ \downarrow & & \downarrow \\ \mathbf{K}_+\,\Gamma[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RV}[*] \end{array}$$

where all the arrows are monomorphisms. The map

$$\mathbf{K}_+\,\mathrm{RES}[*] \times \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*]$$

determined by the assignment

$$([(U, f)], [I]) \longmapsto [(U \times I^\sharp, f \times \mathrm{id})] \tag{3.3}$$

is well-defined and is clearly $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-bilinear. Hence it induces a $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-linear map

$$\Psi : \mathbf{K}_+\,\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*], \tag{3.4}$$

which is a homomorphism of graded semirings. By the universal mapping property, groupifying a tensor product in the category of $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-semimodules is, up to isomorphism, the same as taking the corresponding tensor product in the category of $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$-modules; the groupification of Ψ is still denoted by Ψ. Similarly, there is a $\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]$-linear map

$$\mu\Psi : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\mu\Gamma[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}[*]. \tag{3.5}$$

We shall abbreviate $\otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]}$, $\otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]}$ both as "$\otimes$" below when no confusion can arise.

Proposition 3.18 ([8, Corollary 10.3, Proposition 10.10(1)])**.** *Ψ and $\mu\Psi$ are both isomorphisms of graded semirings.*

Formalization review T001: finite coefficients and the fixed ground. The injectivity assertion above needs review under the present parameter and Γ-morphism conventions. The following obstruction already occurs for a VF-generated henselian ground with nontrivial value group. Take $\mathbb{S} = \mathbb{C}((t))$, normalize $\mathrm{val}(t) = 1$, and put

$$\delta = \tfrac{1}{5}, \qquad R = \{r \in \mathrm{RV} : r^5 = \mathrm{rv}(t)\}, \qquad I = (0,\infty), \quad J = (\delta,\infty).$$

Here R is a five-element object of RES[0] and every $r \in R$ has value δ. The original-ground definable bijection

$$R \times I^\sharp \longrightarrow R \times J^\sharp, \qquad (r,x) \longmapsto (r, rx)$$

has inverse $(r,y) \mapsto (r, r^{-1}y)$. Thus $\Psi([R]_0 \otimes [I]_1) = \Psi([R]_0 \otimes [J]_1)$. The two tensors are nevertheless distinct with the stated balancing ring.

A detector for their difference. All definable finite endpoints in Γ are rational. Define $F : \mathbb{Q} \longrightarrow \mathbb{Z}$ by $F(a) = 1$ on $\frac{1}{5} + \mathbb{Z}$, $F(a) = -1$ on $-\frac{1}{5} + \mathbb{Z}$, and $F(a) = 0$ otherwise; set $F(\pm\infty) = 0$. It is odd and $\mathbb{Z}$-periodic. Give each interval with endpoints $a < b$ weight $F(a) - F(b)$, independently of endpoint inclusion, and each point weight zero. Add the weights over finite disjoint unions to obtain ν. Cutting an interval at c preserves its weight because $(F(a) - F(c)) + (F(c) - F(b)) = F(a) - F(b)$. Translations by integers and reflections $a \mapsto -a + z$, $z \in \mathbb{Z}$, also preserve the weight. Consequently ν is an additive invariant of Γ[1]: these are exactly the permitted affine pieces in this ground by Remark 3.15. In particular, $\nu(I) = 0$, $\nu(J) = 1$, and ν vanishes on finite objects.

Let $D = \mathbb{Z}[\varepsilon]/(\varepsilon^2)$, graded by $\deg \varepsilon = 1$. There are graded semiring homomorphisms

$$\mathbf{K}_+ \mathrm{RES}[*] \longrightarrow D, \quad [X]_0 \longmapsto |X|, \quad [X]_k \longmapsto 0 \ (k > 0),$$

$$\mathbf{K}_+ \Gamma[*] \longrightarrow D, \quad [X]_0 \longmapsto |X|, \quad [X]_1 \longmapsto \nu(X)\varepsilon, \quad [X]_k \longmapsto 0 \ (k \geq 2).$$

They agree on $\mathbf{K}_+ \Gamma^{\mathrm{fin}}[*]$, so induce a homomorphism from the tensor source of Ψ. It sends the displayed tensors to 0 and 5ε, respectively. This proves their inequality, also after groupification. The finite-Γ balancing relations therefore do not suffice to descend this coefficient-dependent translation.

Precise unresolved point. The issue is descent of the RV lift over the original ground. The corresponding step to check in the cited HK proof is the passage from a definable bijection of Γ pieces in Proposition 10.2, Claim 3, to equality of their tensor classes. The latter requires a Γ-morphism, with its lift over the original ground. If this obstruction is valid in HK's stated context, it challenges that argument too; it is not merely a missing convention in the present paper. No verified replacement theorem or repair by changing the ground or coefficients has been established. The earlier phrase "until the intended ground or coefficient conventions are corrected" is withdrawn as premature. The working manuscript remains unchanged. The concrete model and the complete detector above have not yet been constructed in Lean.

Note that, however, [8, Proposition 10.10(2)] does not hold. One of the main motivations of this paper is to construct an alternative to be used in applications.

Notation 3.19. For simplicity, we often drop the constant volume form 0 from the notation. For instance, if $\boldsymbol{U}$ is an object of RV[∗] then it may also denote the object $(\boldsymbol{U}, 0)$ of μRV[∗] with the constant volume form 0.

Notation 3.20. Let $[1] \in \mathbf{K}_+ \mathrm{RES}[1]$ be the class of the singleton $\{1\}$. The class of the singleton $\{1\}$ in $\mathbf{K}_+ \mathrm{RES}[0]$ is the multiplicative identity of $\mathbf{K}_+ \mathrm{RES}[*]$ and hence is simply denoted by 1. We have $[\mathrm{RV}^{\circ\circ}_\infty] = [\mathrm{RV}^{\circ\circ}] + 1$ in $\mathbf{K}_+ \mathrm{RV}[\leq 1]$.

Let I_{sp} be the (nonhomogeneous) semiring congruence relation on $\mathbf{K}_+ \mathrm{RV}[*]$ generated by the pair $([1], [\mathrm{RV}^{\circ\circ}_\infty])$. Let

$$\boldsymbol{P} = [1] - [\mathrm{RV}^{\circ\circ}] \in \mathbf{K}\,\mathrm{RV}[1].$$

The corresponding principal ideal of $\mathbf{K}\,\mathrm{RV}[*]$ is thus generated by the element $\boldsymbol{P} - 1$.